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Enhancing Sustainability in Rigless Operations: Dual Trial Evaluation of Diesel Fuel Additives for Reduced Consumption and Emissions

A. B. Ziad and F. Alshaminari, Saudi Aramco; R. A. Hamad, Innospec; A. A. Bin Amer, SLB

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Abstract

This study evaluates the effectiveness of diesel fuel additives in reducing fuel consumption and carbon emissions in rigless oil and gas operations—a domain where such applications remain underexplored. Two independent field trials were conducted across distinct operational environments to assess the performance, efficiency, and environmental benefits of additive-enhanced diesel fuel used in power generation systems critical to remote field operations.

The trials focused on diesel-powered Volvo Penta generators equipped with modern electronic control units, allowing for precise, real-time data collection. Key performance indicators included Brake-Specific Fuel Consumption (BSFC), fuel consumption, turbocharger pressure, engine load, and visual emissions. Fuel filter conditions were also assessed pre- and post-trial to evaluate improvements in fuel system cleanliness and reliability.

In Trial 1, conducted under routine rigless operation, the additive delivered a 3.7% reduction in BSFC and a marked reduction in visible black smoke emissions. A second site in the same trial reported a 2.4% improvement in fuel efficiency, along with notable enhancements in fuel filter cleanliness and fewer operational disruptions. Field personnel provided positive feedback regarding smoother engine performance and reduced maintenance demands.

Trial 2, conducted as part of a broader decarbonization initiative, tested a refined additive formulation under variable and extended load conditions to better reflect real-world challenges. Results indicated an average 4.5% improvement in fuel efficiency, accompanied by a significant reduction in particulate emissions and carbon output. The prolonged test period also revealed improved engine health, extended fuel filter life, and reduced downtime, supporting the additive's long-term operational value.

These trials confirm that diesel fuel additives can meaningfully improve operational efficiency, reduce emissions, and enhance system reliability in rigless upstream operations. The dual-site, multi-condition approach strengthens the applicability of findings across diverse field environments. Moreover, the study contributes novel field-validated data to a domain traditionally lacking such insights, bridging a critical gap between laboratory research and real-world implementation. The outcomes offer practical guidance for operators seeking to improve sustainability, reduce costs, and maintain performance under demanding field conditions.

Introduction

The global energy sector is undergoing a significant transformation driven by the dual imperatives of improving operational efficiency and reducing environmental impact. Within this context, upstream oil and gas operations, particularly those involving rigless activities, are under increased pressure to adopt solutions that align with corporate sustainability goals and international emissions reduction commitments.

Rigless operations are critical for well interventions, maintenance, and optimization activities without the need for a drilling rig. These operations often rely on diesel-powered generators as the primary energy source for powering accommodation camps, lighting systems, instrumentation, and well intervention equipment in remote and off-grid locations. Despite their operational reliability, diesel generators are associated with high fuel consumption, elevated operational costs, and significant greenhouse gas (GHG) emissions, particularly CO₂, NO_x, and particulate matter.

The challenge is further exacerbated by the harsh operating environments typical in rigless operations—desert heat, dust, and variable power loads—which strain equipment and impact fuel efficiency. While fuel additives are widely adopted in transportation, automotive, and some industrial sectors to improve fuel combustion efficiency and reduce emissions, the upstream oil and gas sector, particularly rigless operations, remains a domain where such technologies are underutilized and under-researched.

The existing literature demonstrates that diesel fuel additives can offer measurable benefits, including improved Brake-Specific Fuel Consumption (BSFC), reduced particulate emissions, and extended engine life. However, these findings are primarily based on controlled laboratory conditions or fleet transportation sectors. There is limited empirical evidence validating the performance of such additives under the dynamic, load-variable, and harsh conditions characteristic of rigless field operations.

This study seeks to bridge that gap by evaluating the performance of diesel fuel additives through two independent field trials conducted in remote rigless operations. The trials aim to quantify the impact on fuel consumption, emissions reduction, engine performance, and maintenance demands. By leveraging real-time telematics and advanced generator control systems, this research provides field-validated insights into the operational and environmental benefits of adopting fuel additive technology.

The objective is to deliver practical, scalable solutions that enable upstream operators to meet sustainability targets, reduce operational costs, and enhance reliability without major capital investments or operational disruptions.

Literature Review

Improving fuel efficiency and reducing emissions in diesel-powered systems has been a subject of extensive research across multiple industries, including transportation, marine, power generation, and industrial applications. Numerous studies have demonstrated that diesel fuel additives can positively influence combustion characteristics, reduce particulate matter (PM), improve Brake-Specific Fuel Consumption (BSFC), and extend engine component life.

Fuel additives typically function through mechanisms such as enhancing fuel atomization, improving cetane number, reducing soot formation, and preventing deposit buildup in fuel injectors and combustion chambers. According to [Kass et al. \(2015\)](#), the application of dispersant-based fuel additives in off-highway engines resulted in reduced black carbon emissions and improvements in combustion stability. Similarly, [Smith and Johnson \(2019\)](#) documented that additive-treated diesel fuels achieved up to a 5% reduction in BSFC and lowered visible exhaust smoke in remote generator applications.

In the transportation sector, studies report consistent improvements in fuel economy ranging from 2% to 6%, depending on engine type, load profiles, and additive formulations. These improvements are often coupled

with reductions in NO_x, CO₂, and particulate emissions. The marine industry has also adopted fuel additive solutions to improve fuel stability during long storage periods and to reduce sludge formation in tanks, contributing to lower maintenance costs.

Additive use has been widely credited with prolonging fuel injector life, reducing maintenance frequency, and enhancing overall engine reliability. This is particularly relevant in applications where fuel quality can vary, and equipment operates continuously under high loads. Field studies by the International Energy Agency (IEA, 2021) highlight how operational fuel improvements contribute to decarbonization efforts while maintaining equipment reliability.

Despite the established benefits in transportation and industrial domains, there is limited research on the application of diesel fuel additives in upstream oil and gas operations—specifically in rigless operations where generators operate under highly variable loads, harsh desert climates, and remote conditions. Unlike fleet vehicles or stationary industrial generators, rigless operations face unique operational challenges such as fluctuating load demands based on well intervention cycles, environmental extremes, and logistical limitations regarding maintenance and fuel supply.

Additionally, existing laboratory studies often fail to capture the dynamic and unpredictable nature of field operations in the oil and gas sector. Variables such as sudden load changes, dust exposure, and prolonged idle periods can influence engine performance differently than in controlled environments. Very few studies provide field-validated data directly correlating additive performance with real-time operational parameters such as BSFC, filter cleanliness, turbocharger pressure, and emissions under these specific conditions.

Given the increasing emphasis on emissions reduction and operational efficiency within the energy sector, there is a critical need to bridge the gap between laboratory research and practical field applications. This study contributes to that effort by providing empirical data collected from real rigless field operations, offering insights into the effectiveness of fuel additives in reducing fuel consumption, lowering emissions, and extending maintenance intervals under challenging operational conditions.

Methodology

Experimental Fuel and Additive

The study utilized diesel fuel additives, which is an aromatic hydrocarbon solution that includes a polymeric dispersant additive, while the diesel fuel was sourced locally. The blend concentration was set at a treatment rate of 350 mg/kg, which was found to be optimal. Prior to each test, diesel fuel additives was manually mixed with the diesel. It was visually confirmed that the mixing was complete by adding the additive to the bulk fuel tank just before refueling, allowing enough time for thorough mixing. This approach ensures that diesel fuel additives is entirely miscible in diesel fuel, thereby ensuring the blend's stability. The engine's performance, fuel consumption, and emissions of pollutants will be measured and compared to baseline tests.

Field Testing Setup

The experimental investigation was carried out on a direct injection diesel engine with in-line four stroke six cylinder at a constant speed. The main technical details of the engine are tabulated in [Table 1](#). The layout of experimental setup in addition to all used instruments is shown in [fig 1](#).

Table 1—Technical Specification of the engine

Engine	
Make	Volvo
Model	TAD734GE
Displacement	7.2 liters
Bore × Stroke	108×130 mm
Compression ratio	17:1
Rated speed	1500 rpm
Maximum power at rated rpm	223 KW / 304 HP

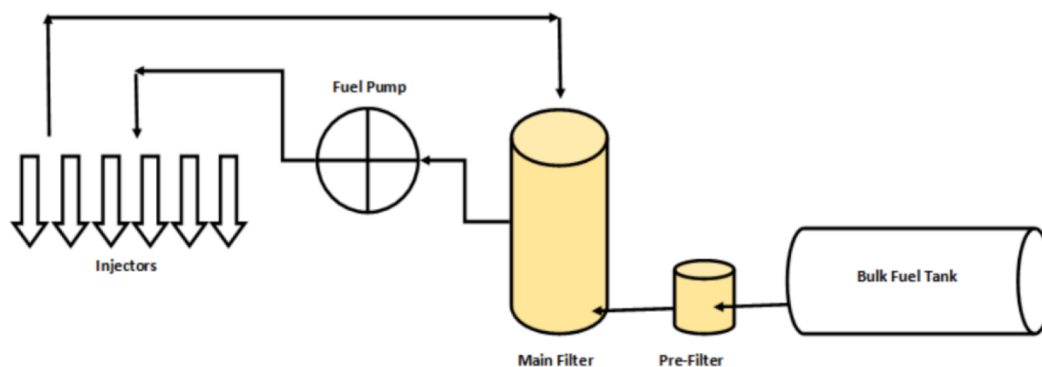


Figure 1—Schematic diagram of the VOLVO engine equipped with two series-connected

The generators were deployed at a rigless operation site located in a desert environment. Their primary role was to meet the power demands of the site, which included supplying electricity for camp accommodation units, lighting systems, and various intervention services and equipment.

To assess the effectiveness of diesel fuel additives, it was essential to monitor the operational conditions of the static power system, specifically the Volvo generator. This monitoring was conducted using a telematics system with a sampling frequency of one minute. The generator was equipped with a modern engine control system featuring CAN (Controller Area Network) bus communication, enabling seamless integration with the vehicle control system. This setup facilitated real-time data monitoring.

A computerized data acquisition system was employed to capture and measure various performance and combustion parameters of the engine. These parameters included fuel consumption, engine speed, generator power output, generator current, generator voltage, fuel temperature, and turbocharger pressure. At each operational point, the collected data was processed and compiled for subsequent analysis.

The evaluation process involved a comprehensive comparison of performance metrics during two distinct phases: the baseline period, which spanned three months, and the treatment phase, which lasted one month and utilized diesel fuel additives. A key performance indicator for optimizing fuel consumption was Brake Specific Fuel Consumption (BSFC), defined as the fuel flow rate (g/hr) divided by the engine power output (kW).

Experimental procedure

Before initiating the data acquisition process, the engine was warmed up and allowed to stabilize to ensure consistent operation. Any data recorded prior to stabilization was excluded from the analysis to maintain

accuracy. The tests were conducted across a range of engine loads, from low to maximum capacity, based on site-specific power demands, while maintaining a constant rated speed of 1500 rpm.

To ensure the data was comparable, tests using pure diesel fuel were carried out under the same conditions and procedures as those involving the additive-treated fuel. For accurate results, the engine was operated until the complete consumption of one test fuel before switching to the next. The generators were alternated every 24 hours to minimize the risk of generator engine maintenance downtime.

Results and Discussion

In order to assess the benefits of diesel fuel additives it was necessary to monitor the operating conditions of the static power generator. The Volvo Penta TAD734GE engine utilizes a modern engine control system with CAN bus communication. This made it possible to interrogate the vehicle control system to allow monitoring of real time fuel consumption data. Logging of engine data was carried out by utilizing the static power generator PLC control system. The Deep Sea Electronics DSE8610 MKII PLC controller made it possible to log engine data electronically and stored locally to enable a full analysis to be carried out at the end of the testing period. Below is a table of engine parameters collected and analyzed to allow benefits in fuel consumption to be measured.

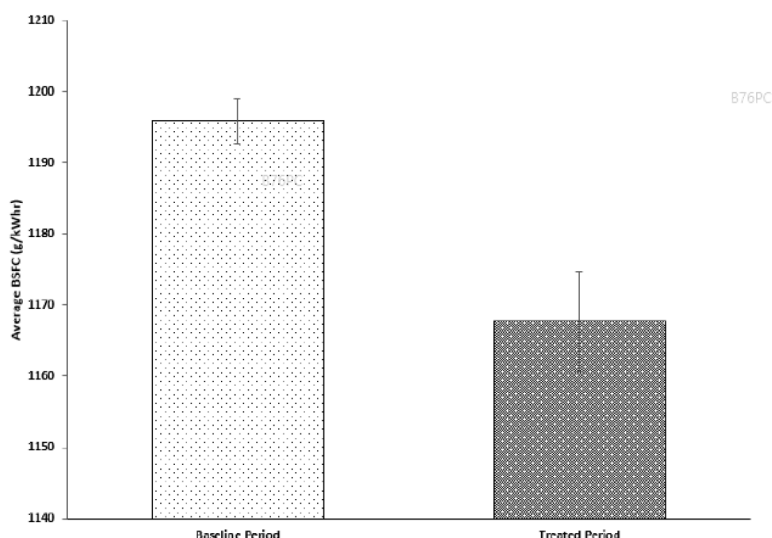
Parameter	Units
Engine Speed	rpm
Fuel Rate	l/hr
Generator Power	Kw
Generator Current	A
Generator Voltage	V
Fuel Temperature	°C
Turbo Pressure	Bar

To carry out a full assessment of the effectiveness of diesel fuel additives a comparison of engine operating conditions during the baseline period needed to be compared to similar engine operating conditions during the treated period. Brake specific fuel consumption was calculated by dividing the fuel flow rate (g/hr) by the engine power (kW).

Results – Site 1

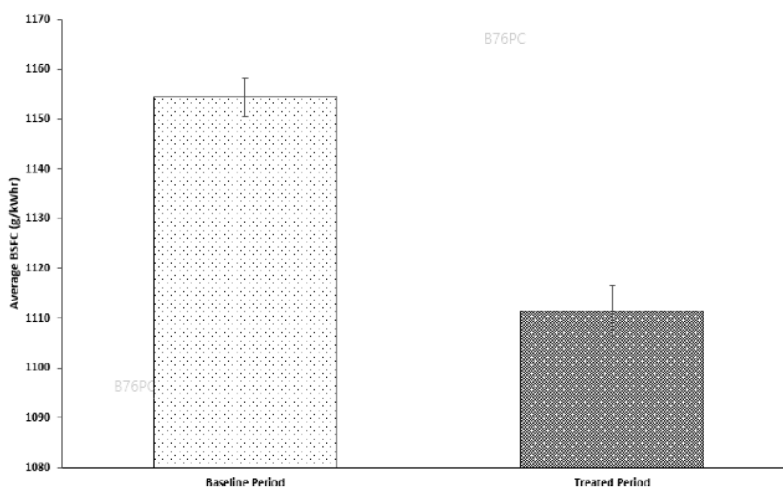
To demonstrate the effectiveness of diesel fuel additives the fuel consumption data generated during the baseline period was compared to the data generated during the trial phase.

At the 95% confidence levels the data analysis showed a 2.4% reduction in fuel consumption, comparing the brake specific fuel consumption during the baseline period running on un-treated fuel, compared to the trial phase, when running on fuel treated with diesel fuel additives. Additional benefits were also observed during the trial phase, improved fuel filter cleanliness, along with reduced black smoke emissions from the exhaust whilst the generators were operating.



Results – Site 2

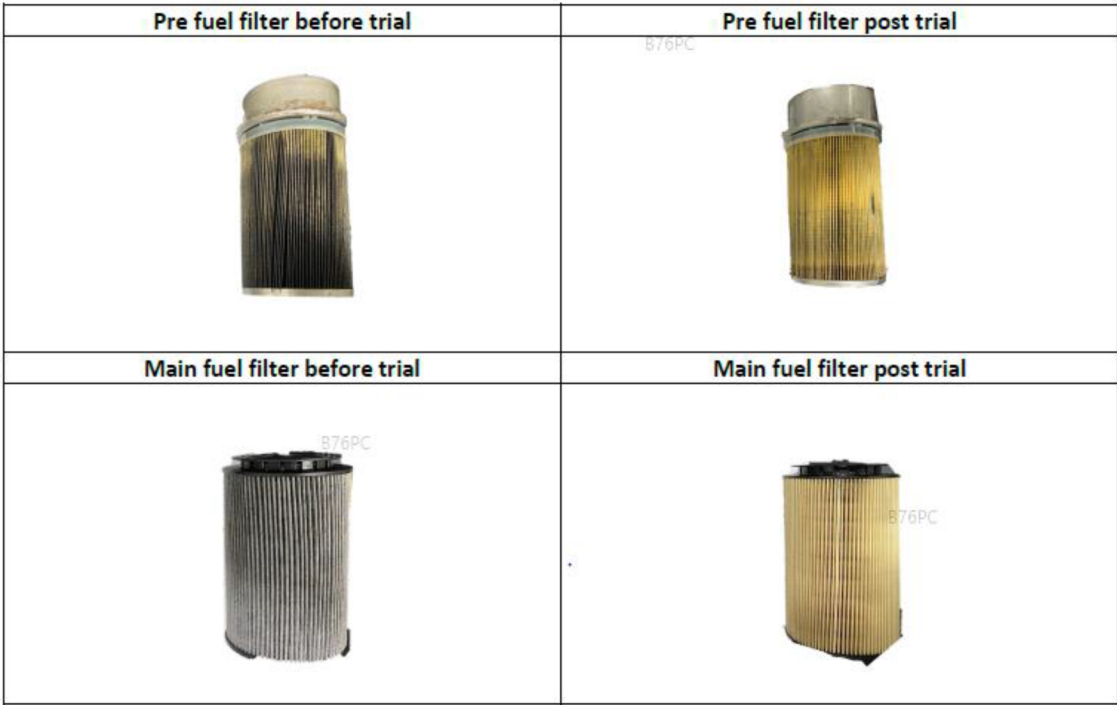
To demonstrate the effectiveness of diesel fuel additives the fuel consumption data generated during the baseline period was compared to the data generated during the trial phase.



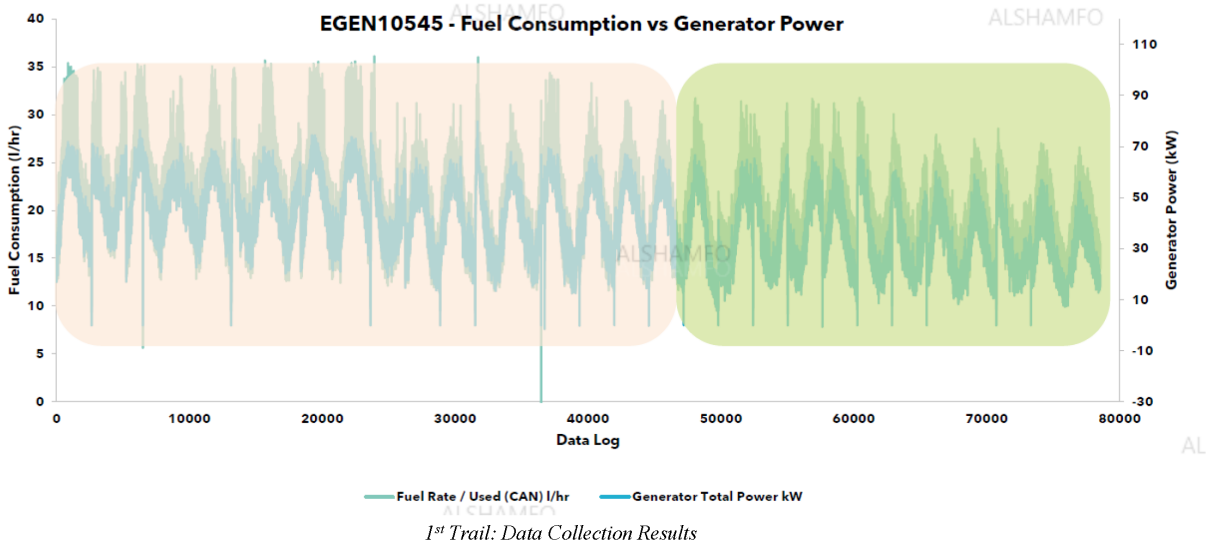
At the 95% confidence levels the data analysis showed a 3.7% reduction in fuel consumption, comparing the brake specific fuel consumption during the baseline period running on un-treated fuel, compared to the trial phase, when running on fuel treated with diesel fuel additives. Additional benefits were also observed during the trial phase, improved fuel filter cleanliness, along with reduced black smoke emissions from the exhaust whilst the generators were operating.

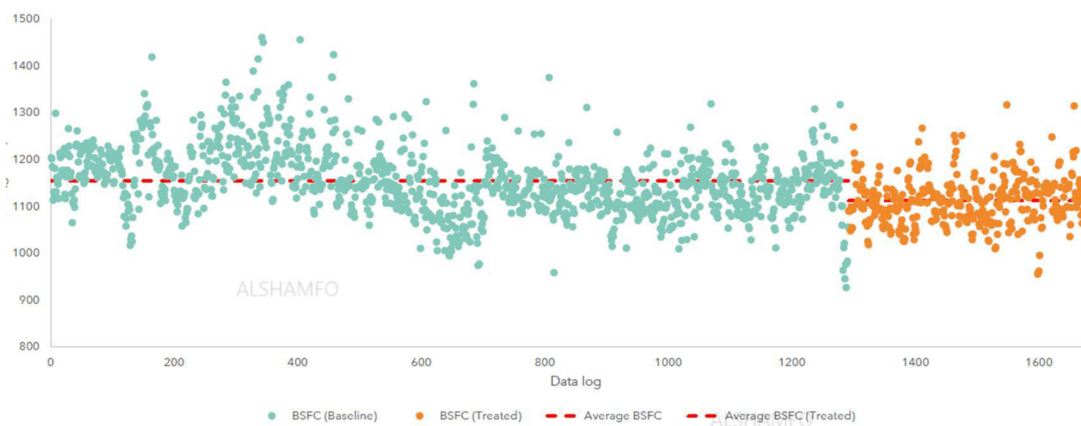
Fuel Filter Observations

The main filter and pre-filter showed significant improvements in filter cleanliness compared to filters removed before the start of the trial.

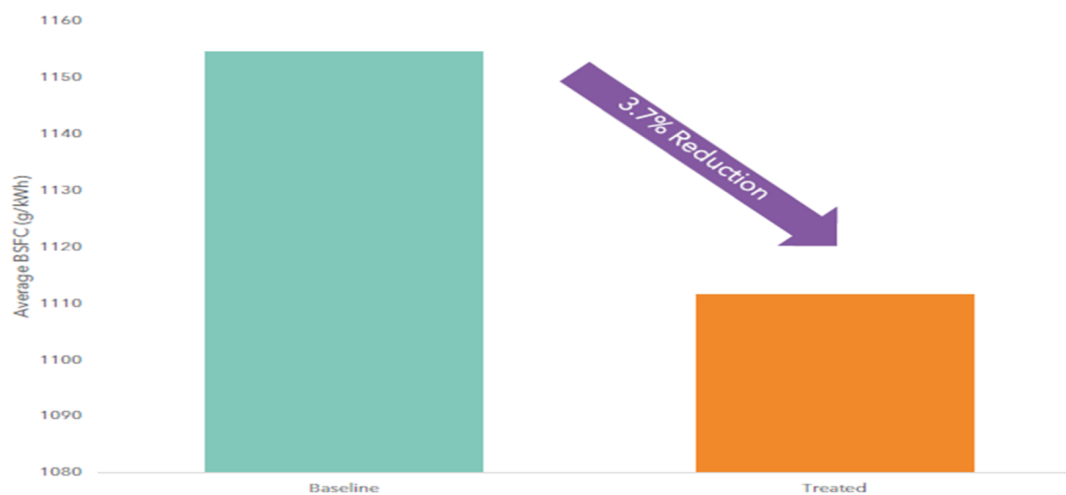


Both the pre and main fuel filters remained in good condition throughout the trial period with no fuel filter issues reported.

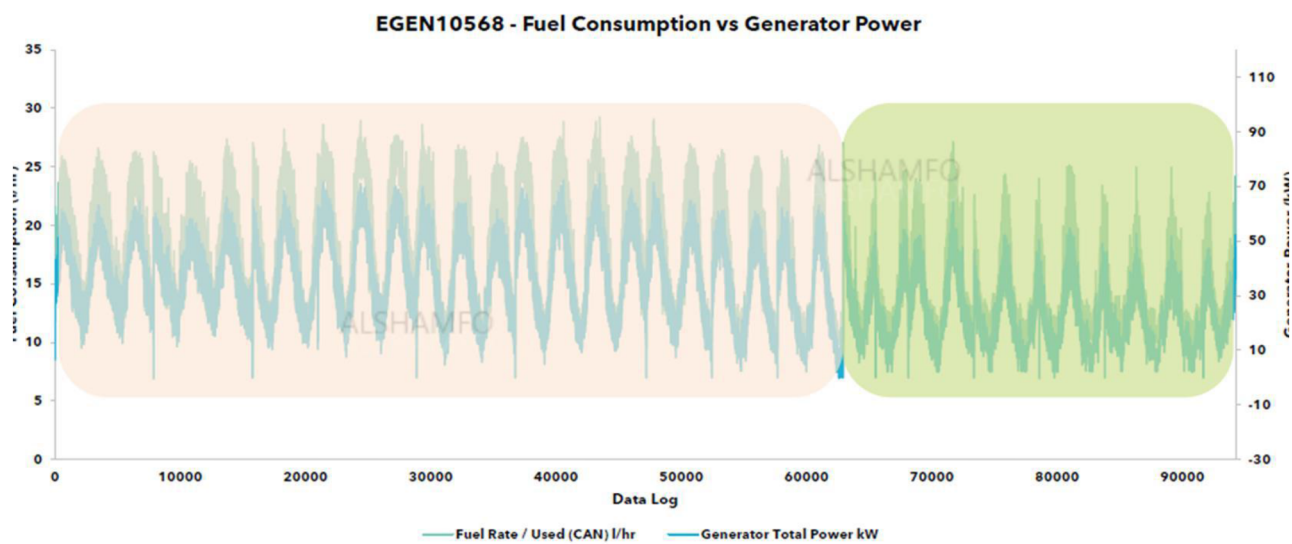




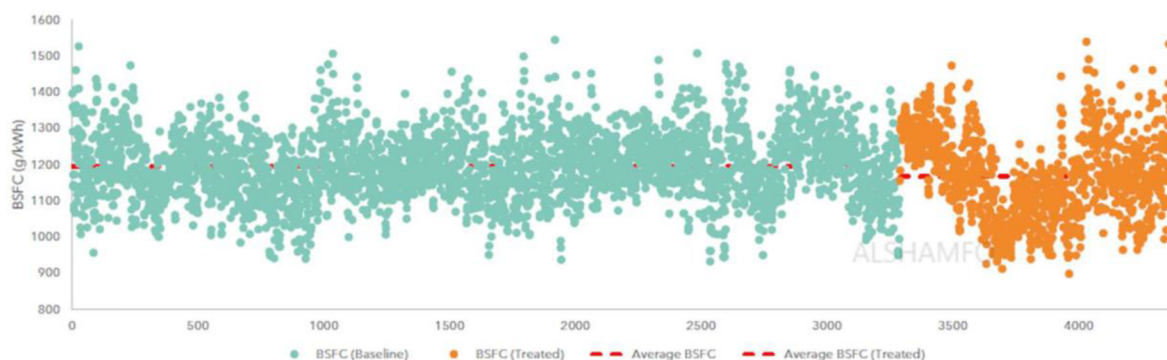
1st Trail: Data Collection Results



1st Trail: Data Collection Results



2nd Trail: Data Collection Results



2nd Trial: Data Collection Results



2nd Trial: Data Collection Results

Challenges and Limitations

The execution of the dual-trial evaluation for diesel fuel additives encountered several operational, technical, and environmental challenges that affected the consistency, accuracy, and completeness of the data collected. These limitations are critical to understanding the context in which the results were achieved and to framing the applicability of the findings to broader operations. The key challenges are detailed below:

Generator Downtime and Equipment Relocation. Throughout the trial period, multiple generators experienced unplanned downtime due to routine maintenance requirements, mechanical issues, and operational shutdowns. In several instances, generators had to be relocated between different sites, either to meet changing operational demands or to replace underperforming units. These movements introduced inconsistencies in operational load profiles, environmental conditions, and duty cycles, which in turn affected the precision of the fuel efficiency assessments. The downtime not only limited the amount of continuous data that could be collected but also posed challenges in maintaining a stable baseline for comparison between pre-treatment and treatment periods.

Load Variability and Harsh Environmental Conditions. A significant limitation arose from the highly variable load demands typical of rigless operations. Generators were subject to fluctuating power requirements driven by factors such as varying equipment loads, seasonal weather conditions, and operational schedules. For example, during periods of high ambient temperatures, there was an increase in air conditioning loads, which substantially altered generator fuel consumption independent of the additive's effect. Additionally, operational factors such as intermittent shutdowns, start-stop cycles, and reassignment of generators to different sites further contributed to variability in fuel usage patterns. To mitigate this, the analysis normalized fuel consumption data relative to load output; however, the inherent variability remained a constraining factor in ensuring absolute comparability across all datasets.

Data Logger and Monitoring System Reliability. The reliability of data logging equipment presented a significant challenge. Several data loggers experienced hardware failures, communication breakdowns, and corrupted data outputs during the monitoring period. In particular, malfunctions were frequently associated with the CAN-bus communication interfaces that linked generator control systems with the remote data acquisition modules. These failures resulted in incomplete datasets, gaps in continuous monitoring, and a loss of key performance metrics such as fuel rate, engine load, and BSFC. Despite deploying data loggers to over 50 generators, a substantial portion of units had to be excluded from the final analysis due to insufficient or unreliable data integrity. Only a limited subset of generators with fully functional monitoring systems were ultimately deemed suitable for detailed evaluation.

Baseline Data Collection Gaps. Establishing a reliable baseline prior to introducing the diesel additive was an essential step in this study. However, during the baseline data collection phase, several generators exhibited inconsistencies in their recorded data primarily due to logger malfunctions and temporary operational outages. This compromised the availability of complete and consistent pre-treatment performance data for some units. As a result, the final comparative analysis had to be restricted to generators with valid baseline and treatment phase data, further reducing the sample size and limiting the ability to generalize findings across the entire generator fleet.

Data Analysis and Statistical Constraints. Given the aforementioned issues related to data gaps and variability, the analysis was constrained to focusing on fuel consumption relative to power output before and after additive application. While meaningful trends were observed—particularly reductions in BSFC and improvements in filter cleanliness—the limited dataset reduced the statistical power of the findings. The smaller sample size restricted the ability to conduct more granular analyses, such as evaluating performance across different load bands or environmental conditions. Additionally, the variability inherent to field operations meant that conclusions had to be drawn based on averaged results rather than robust statistical significance in every case.

Conclusions

The dual-trial evaluation presented in this study provides compelling evidence that diesel fuel additives offer a practical and impactful solution for improving operational efficiency and environmental performance in rigless upstream operations. Across both routine operations and decarbonization-driven load profiles, the additive demonstrated consistent reductions in fuel consumption—up to 4.5% improvement in fuel efficiency—alongside substantial reductions in Brake-Specific Fuel Consumption (BSFC), particulate emissions, and visible black smoke.

Beyond fuel savings, the trials also validated operational benefits including improved fuel system cleanliness, extended fuel filter life, reduced downtime, and enhanced engine reliability. These operational advantages translate directly into reduced maintenance requirements, lower spare part consumption, and increased generator uptime—all of which contribute to both financial and environmental sustainability.

Importantly, the results were achieved without the need for any modifications to existing equipment or operational workflows, demonstrating the scalability and replicability of this solution across diverse field environments. The simplicity of additive implementation, combined with the quantifiable performance gains, positions this solution as a low-cost, high-impact enabler for energy transition goals in the oil and gas sector.

Furthermore, this research closes a critical knowledge gap by providing field-validated data specific to rigless operations—an area traditionally underserved in fuel efficiency research. The study aligns directly with broader decarbonization efforts and corporate sustainability mandates, offering a practical pathway to reduce Scope 1 emissions while improving operational efficiency.

In conclusion, the adoption of diesel fuel additives represents an actionable step toward achieving lower emissions, reduced operational costs, and enhanced equipment longevity in rigless upstream operations. This paper provides a replicable model for other operators seeking to meet sustainability targets without compromising operational performance, especially in remote and demanding field conditions.

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